

Lab 1: Set Up the Things

Handout: 29 Sep 2026 Hand-in: 6 Oct 2026, eDimension

Overview

In this lab, your group will set up a **MakePython ESP32** development board and blink an external LED using your own MicroPython program.

- Use the **MakePython ESP32 board (without a camera)**, one LED, a 330 ohm resistor, a breadboard, jumper wires, and a USB data cable.
- Connect the LED to **3V3** and **IO5** as shown below. Disconnect USB power before changing the wiring. The resistor must be in series with the LED.



- Install **Thonny** on Windows or macOS. Select **MicroPython (ESP32)** as the interpreter and choose the board's serial port. A `>>>` prompt in the Shell confirms that MicroPython is ready.
- Set **Pin(5, Pin.OUT)** high and low every 0.5 seconds. With this wiring, **0 turns the LED on** and **1 turns it off**. Run the program and save it on the board as **main.py**.

Hints:

1. If your computer cannot find the port, first try a different USB **data** cable. On Windows, look for a CP210x COM port in Device Manager; on macOS, look for a new `/dev/cu.*` port. Install the Silicon Labs CP210x driver only if the port remains absent.
2. If Thonny does not show `>>>`, click Stop and confirm the selected port. Install or update **ESP32 / WROOM** MicroPython through Thonny only if firmware is missing. A separate firmware `.bin` file is not required for a board that already opens the Shell.
3. If the LED does not blink, check the LED leg direction, breadboard rows, resistor, and the **3V3** and **IO5** labels. The board's steady power indicator is not the external LED for this task.

What to hand in?

1. **Demonstrate your setup with photos or a short video.** Include it with your report or provide a working link if the file is too large. Show:

- The development environment is installed and connected to the ESP32 (for example, Thonny showing the MicroPython Shell).
- The external LED blinks as expected when your code runs.

Hardware for this lab: use the MakePython ESP32 development board and the external LED circuit described on page 1.

2. **Write a brief lab report** documenting:

- Any issues encountered during setup and how your group solved them. If there were no issues, say so briefly.
- Your group ID and all members' names and student IDs, including audit students if any.

3. **Identify the functions of the components and devices you used.** Label them briefly on an image of your setup (for example, the board, IO5 and 3V3 pins, resistor, LED, and USB connection).

Submit **one lab report per group** to eDimension by **Tuesday, 6 October 2026**. The 2026 syllabus specifies the hand-in date but not a cut-off time; follow the assignment time shown on eDimension.

Total marks for this lab: 25pt

- **10pt: Environment set-ups** - show that Thonny connects to the ESP32 and opens the MicroPython Shell.
- **10pt: Blink the LED** - demonstrate an external LED blinking under control of your program.
- **5pt: Device overview** - label the components and briefly explain their functions.